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Roll No

CE-702(B)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Environmental Engineering-II

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Describe, for a client, the four types of sedimentation systems and the reasons you would recommend one over another. 7
b) If the 5 days BOD of a sample is 276 mg/l and ultimate BOD at the same temperature is 380 mg/l, at what rate the waste is oxidized? 7
2. a) A waste water treatment plant with a design flow of 50,000 m³/d has influent BOD 250 mg/l and effluent BOD of 10 mg/l. The Y (decimal fraction of food converted in kg to biomass kg) is 0.45. Endogenous decay coefficient $k_d = 0.05 \text{ d}^{-1}$. Assumptions (MLSS = 2000 mg/l, under flow concentration from secondary clarifier = 9500 mg/l). Design a completely mixed activated sludge reactor, determine: 7
 - i) Volume of reactor
 - ii) Mass and Volume of solids that must be wasted each day
 - iii) Recycle ratio.

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- b) Explain the process of the formation and sloughing of the slime layer in trickling filters. Draw a neat diagram of high-rate trickling filter. 7
- 3. a) Highlight the advantages and disadvantages of 'Ultra filtration' and 'Micro filtration'. 7
- b) Explain the need for removal of nitrogen from waste water. Also, explain the process of Nitrification and Denitrification for nitrogen removal. 7
- 4. a) What are the six "criteria" air pollutants? Which one occurs most commonly during summer-time? Why? 7
- b) Write a short note on photo-chemical smog. (350 words) 7
- 5. a) With a neat sketch Explain the Plume behavior for the different atmospheric conditions. 7
- b) Determine the stack gas temperature for the following data: 7
 - i) Physical height of stack = 90 m
 - ii) Wind velocity at the top of stack = 2.5 m/s
 - iii) Pressure = 1000 millibars
 - iv) Stack gas velocity at the top of stack = 11 m/s
 - v) Effective height of stack = 98.5 m
 - vi) Ambient temperature = 200C
 - vii) Inside diameter of stack = 0.9 m
- 6. a) A water treatment plant has two clarifiers treating 0.2 m³/s of water. Each clarifier is 5m wide, 25 m long and 5 m deep. Determine the percent of particles retained in the clarifier. 7

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- b) Determine the size of a standard rate trickling filter for the following data: 7
- i) Sewage flow = 7 MLD
 - ii) BOD of raw sewage = 300 mg/l
 - iii) BOD removed in Primary Clarifier = 40%
 - iv) Final effluent BOD = 30 mg/l
 - v) Depth of filter = 1.5 m
7. A factory emits 20 g/s of SO_2 at height H. The wind speed is 3 m/s. At a distance of 1 km downwind, the values of σ_y and σ_z are 30 m and 20 m, respectively. What are the SO_2 concentrations at the centerline of the plume and at a point 60 m to the side of and 20 m below the centerline? 14
8. A city discharges $1.25 \text{ m}^3/\text{s}$ of waste water into a stream whose minimum rate of flow is $8.0 \text{ m}^3/\text{s}$. The velocity of the stream is about 3.0 km/h. The temperature of the waste water is 20°C and that of the stream is 15°C . The 20°C BOD5 of the waste water is 250 mg/L and that of the stream is 2 mg/L. The waste water contains no dissolved oxygen, but the stream is flowing with saturated DO concentration of 9.2 mg/L. Saturated DO at 15°C is 10.2 mg/L. At 20°C , de-oxygenation constant (K_d) is estimated to be 0.3 per day and re-aeration constant (K_a) is 0.7 per day. Determine the critical oxygen deficit and its location. Also estimate the 20°C BOD5 of a sample taken at the critical point. Use the temperature coefficients of (θ) 1.135 for k_d and 1.024 for k_a . 14
